HOMEWORK ASSIGNMENT 1 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
  - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
- Solutions to this assignment will be released the morning of Thursday, August 27, to give you plenty of time to review them before the corresponding quiz on Monday August 31

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1. **Variable Classification.** Classify each of the following variables as either categorical or quantitative.

   (a) The amount of time (in minutes) it takes a student to get to Cal Poly from their place of residence.

   ◯ Categorical √ **Quantitative**

   Solution: Quantitative: ordinary arithmetic operations on the values has interpretive meaning (1 minute + 2 minutes = 3 minutes). Said differently, the values actually *represent* numbers - 3 in this context really means 3, and is not a placeholder for some other value.

   (b) The mode of transportation (car, bike, bus, etc.) a student uses to get to Cal Poly from their place of residence.

   √ **Categorical** ◯ Quantitative

   Solution: Categorical: observations fall into one of a set of fixed levels (car, bike, bus, etc.), and ordinary arithmetic operations do not make sense (what does "bike" + "car" equal?)

   (c) The proportion of the day a Cal Poly student spends on their phone.

   ◯ Categorical √ **Quantitative**

   Solution: Quantitative: ordinary arithmetic operations on the values has interpretive meaning (2% + 3% = 5%). Said differently, the values actually *represent* numbers - 5 in this context really means 5, and is not a placeholder for some other value.

   (d) The number of classes a Cal Poly student is enrolled in this semester.

   ◯ Categorical √ **Quantitative**

   Solution: Quantitative: ordinary arithmetic operations on the values has interpretive meaning (1 class + 2 classes = 3 classes). Said differently, the values actually *represent* numbers - "3" in this context really means 3, and is not a placeholder for some other value.

   (e) A Cal Poly student's favorite coffee shop in town

   √ **Categorical** ◯ Quantitative

   Solution: Categorical: observations fall into one of a set of fixed levels (SloDoCo, Scout, etc.), and ordinary arithmetic operations do not make sense (what does "SloDoCo" + 'Scout" equal?)

2. **A Picture is Worth a Thousand Words.** For each of the following scenarios, identify the plot or plots that are most appropriate. If there are multiple potentially appropriate plots, please list them all.

   (a) Yuri has collected data on the speeds (in miles per hours) of several cars traveling along Highway 101, and wants to make a plot that displays the distribution of speeds.

   Solution: Yuri wants to make a plot that visualizes the distribution of a single quantitative variable; as such, either a **dotplot** or a **histogram** would be appropriate here.

   (b) Qing has surveyed 100 people and asked them what brand of bottled water they most prefer. She now wants to make a plot that displays the distribution of bottled water brand preferences.

   Solution: Qing wants to make a plot that visualizes the distribution of a single categorical variable; as such, a **barplot/bargraph** is most appropriate.

   (c) Miguel wants to determine whether or not there is a relationship between the amount of sleep students get the night before an exam and their score on the exam. Specifically, he would like to use an appropriate plot to make this determination.

   Solution: Miguel wants to make a plot that visualizes the relationship between two numerical variables; as such, a **scatterplot** is most appropriate.

   (d) Puja has surveyed the students in her class, and asked them which month they were born in. She would like to visualize the distribution of birth months, to determine which months are most popular.

   Solution: Puja wants to make a plot that visualizes the distribution of a single categorical variable; as such, a **barplot/bargraph** is most appropriate.

3. **Living Life to the Fullest.** The *World Bank* is a collection of several agencies, dedicated to studying the global effects of poverty. As a part of their mission, they provide open source data on various attributes of countries around the globe, one of which is a country's average life expectancy. Shown below is a histogram of this variable, as it was observed in 2024:

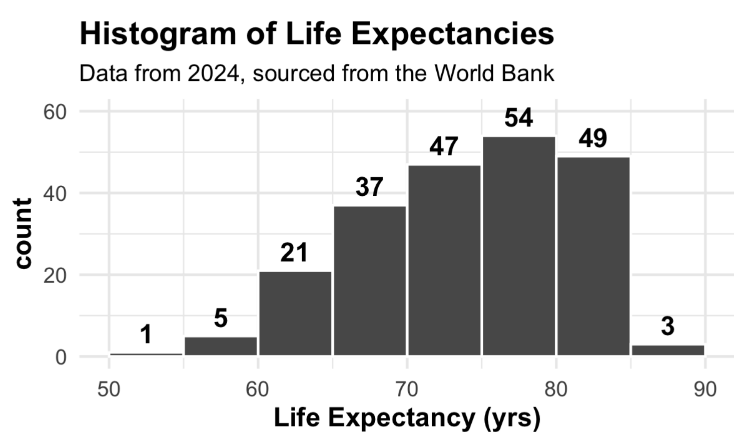


Figure 1: Histogram of Life Expectancy (in years) of 217 countries. Data has been sourced from the World Bank Open Data Repository[1]

Min	54.63
Q_1	68.74
Median	74.71
Mean	74.02
Q_3	79.78
Max	86.50
SD	7.11

[1] https://data.worldbank.org/

(a) What is the observational unit in this dataset?

Solution: An observational unit in this dataset is **one country**. **Note:** recall that on quizzes and exams, this will typically be phrased as "Describe an observational unit," or something similar - as such, you should answer **in singular form** (e.g. one observational unit in this dataset is one country; don't just say "countries").

(b) Describe the distribution of life expectancies. Remember that there are four main characteristics you should describe: shape, center, variability, and outliers. Further recall that your description of "shape" should include mention of the number of modes, symmetry, and skew.

Solution: The shape of the distribution is unimodal, asymmetric, with a left skew. Because the distribution is skewed, we describe its center using the median, which is around 74.71 years, and its variability using the IQR, which is 17.76 years. There are no outliers present (though one might argue that the single country with life expectancy between 50 and 55 years could be considered an outlier).

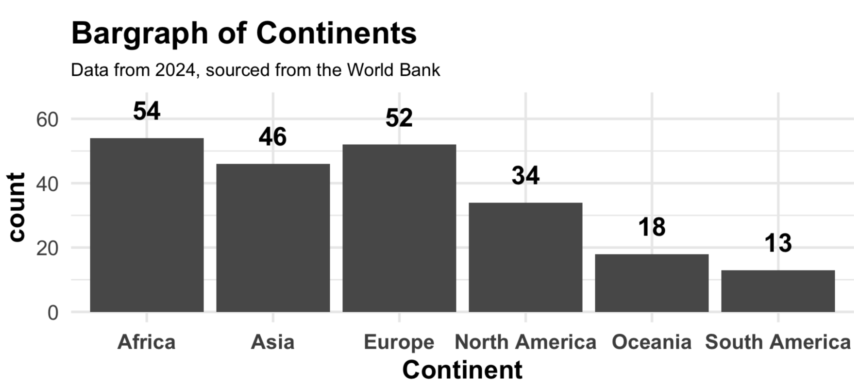
(c) Do you think the median life expectancy is higher, lower, or about the same as the mean life expectancy? Why?

Solution: For left-skewed distributions, the median will be **larger** than the mean. (Intuitively, the tail "pulls" the mean in the direction of the skew.)

(d) What proportion of countries in the dataset have life expectancies between 70 and 80 years?

Solution: From the histogram, the number of countries with life expectancies between 70 and 80 years is 47 + 54 = 101. Since there are a total of 217 countries included in the dataset, the proportion of countries with life expectancies between 70 and 80 years is **101 / 217** $\approx$ **0.4654** = **46.54%**.

(e) Shown below is a bargraph of the continents of the 217 continents included in the dataset.



Does it make sense to describe the shape of this graph (left-skewed, right-skewed)? Why or why not?

Figure 2: Bargraph of Continents of 217 countries.

Solution: **No**, it does not make sense because Skew is a descriptor of a *quantitative* variable and **Continent** is a categorical variable. We can rearrange the order of the categories on a bargraph without changing the meaning of the graph (which is not the case with a histogram) - therefore, even though Figure (2) appears to have a "right-skew," we could simply reverse the order of the continents as they appear on the graph to get a "left-skew," and the plot would still be valid!

4. **Follow the Numbers!** A list of twelve numbers has sample mean equal to 4 and median equal to 4.6.

   (a) **True or False**: the number 4 must have been one of the original twelve numbers.

   ◯ True $\surd$ **False**

   Solution: Recall that we have no guarantee that the mean or median of a list of numbers will be one of the numbers included in the list!

   (b) **True or False**: the number 4.6 must have been one of the original twelve numbers.

   ◯ True $\surd$ **False**

   Solution: For an even number of observations, the median will be the mean of the "middle two" and therefore may not be an element of the original dataset. (For an odd number of observations, the median *will* be one of the original numbers.)

   (c) What was the sum of the twelve numbers? How do you know?

   Solution: We know that the mean is equal to the sum of the numbers divided by the number of elements:

$$(\text{mean}) = \frac{\text{sum of elements}}{\text{total number of elements}}$$

   Multiplying both sides by (total number of elements) reveals that

$$(\text{sum of elements}) = (\text{mean}) \times (\text{total number of elements})$$

   Plugging in $(\text{mean}) = 4$ - which is provided in the original problem statement - and (total number of elements) = 12 - which is also provided in the original problem statement - gives us that the sum of the elements is

$$4 \times 12 = \mathbf{48}$$

   As an Aside: This is very much like Question 9(a) from Lecture Activity 2.

   (d) A new (thirteenth) number is incorporated to the list, and it is equal to 3.25. What is the mean of all thirteen numbers?

   Solution: From part (c), we know that the sum of the original twelve numbers is 48. Incorporating a thirteenth number equal to 3.25 gives us a new sum (of all thirteen numbers) of 48 + 3.25 = 51.25. To find the new mean, we divide this by the new total number of elements (13), which gives a new mean of 51.25 / 13 $\approx$ **3.9423**. **As an Aside:** This is very much like Question 9(b) from Lecture Activity 2.